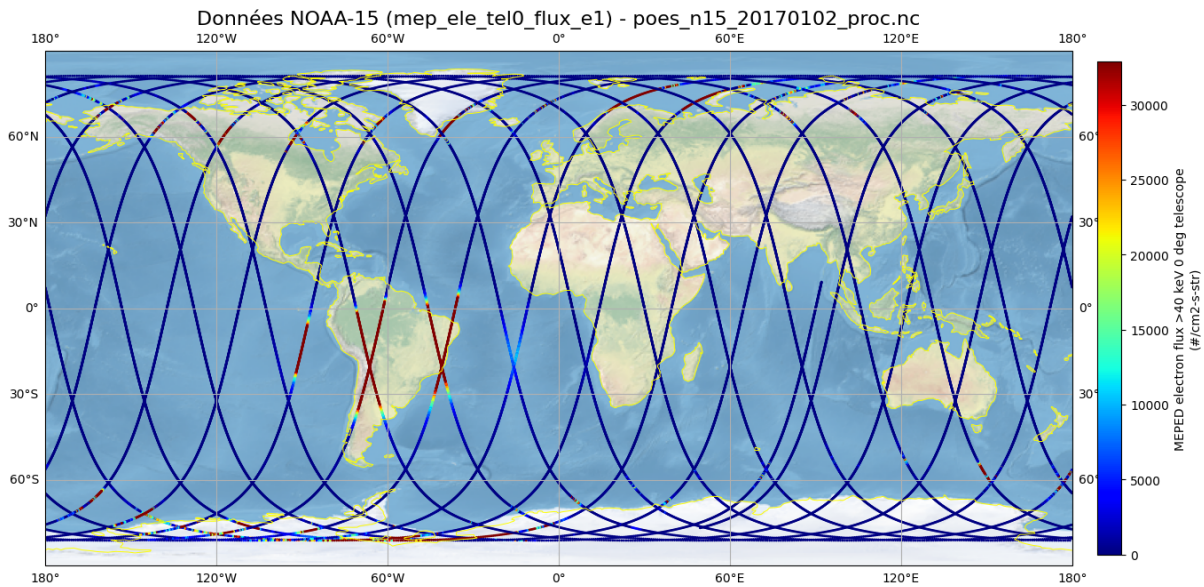
IPSA

Institut Polytechnique des Sciences Avancées

Prediction of 1 MeV Electron Fluxes in Low Earth Orbit

Using NOAA-15 and Solar Wind Data

SP-519: Space Data Processing
Full Project: Steps 1 and 2



Authors:
Casarotto Gabriel
Andre Hippolyte

Date:
December 4, 2025

Academic year 2025–2026

Contents

1 Introduction **1**
1.1 Scientific context . . . 1
1.2 Project objectives . . . 1
1.3 Report structure . . . 1

2 Data sources and preprocessing **1**
2.1 Data sources . . . 1
2.2 NOAA-15 preprocessing pipeline . . . 1
2.3 Analysis of the L–time diagram . . . 2
2.3.1 Key observations . . . 2

3 Statistical analysis **3**
3.1 Flux distribution and event definition . . . 3
3.2 Main visualizations . . . 3

4 Integration of OMNI data and preparation for modelling **3**
4.1 Acquisition and selection of OMNI variables . . . 4
4.2 Cleaning of OMNI data . . . 4
4.2.1 Handling of missing values (NaN) . . . 4
4.2.2 Physical outlier filtering . . . 4
4.3 Aggregation and scaling . . . 4

5 Regression modelling (Flux prediction) **4**
5.1 Features and preparation for regression . . . 4
5.2 Model training per L-shell . . . 5
5.3 Prediction example ($L = 4.0$) . . . 5

6 Results and discussion **5**
6.1 Performance of the regression model . . . 5
6.2 Limitations and perspectives . . . 5
6.2.1 Identified limitations . . . 5
6.2.2 Future improvements . . . 6

7 Conclusion **6**

1 Introduction

This project aims to predict high-energy (1 MeV) electron fluxes in low Earth orbit using data from the NOAA-15 satellite and solar wind parameters. Understanding and forecasting these fluxes is crucial for satellite protection and for managing risks associated with space radiation.

1.1 Scientific context

The Van Allen radiation belts trap energetic particles around the Earth. Solar activity and geomagnetic storms can significantly enhance these fluxes, posing risks to space systems and astronauts.

1.2 Project objectives

- Analyse electron fluxes measured by NOAA-15 over the period 2020–2024.
- Identify high-flux events based on a statistical analysis.
- Develop predictive models (regression and classification).
- Evaluate forecast skill at different time horizons (**+12h, +24h, +36h, +48h**).

1.3 Report structure

This report presents the project in two main parts:

- **Part 1:** Acquisition, preprocessing, and exploratory analysis of NOAA-15 data.
- **Part 2:** Integration of OMNI solar wind data and development of regression and classification models.

2 Data sources and preprocessing

2.1 Data sources

The data used come from two main sources:

- **NOAA-15 SEM-2**: 1 MeV electron fluxes (P6 channel: `mep_pro_tel90_flux_p6`) and geophysical parameters (`L_IGRF`) for the period 2020–2024, in NetCDF format.
- **Solar parameters (OMNI-2)**: solar wind speed (**V**), B_z component of the interplanetary magnetic field (**BZ_GSM**), and solar radio flux index **F10.7**.

2.2 NOAA-15 preprocessing pipeline

The preprocessing is structured in several sequential steps:

1. **Automated download**: Python script to retrieve the NOAA-15 NetCDF files (2020–2024).
2. **Time reconstruction**: combination of variables `year`, `day`, `msec` to obtain an absolute timestamp.
3. **Data cleaning**: replacement of `_FillValue` by `NaN`, removal of zero or negative fluxes, and **statistical outlier filtering** (beyond the 0.1^{st} and 99.9^{th} percentiles).

4. **Physical filtering**: restriction of the L-shell to $1 \leq L \leq 7$ and application of physical limits to OMNI data ($V \in [200, 750]$ km/s, $B_z \in [-30, 30]$ nT, $F10.7 \in [50, 300]$ s.f.u).

5. **Spatial and temporal binning**:

   - Temporal: 12 hour intervals.
   - Magnetic: L-shell rounded to 0.1.

6. **Statistical aggregation**: computation of the mean flux (and other variables) for each pair (date_bin, L_binned).

The final DataFrame contains the columns: `date_bin`, `L_binned`, `flux`, and a time index `t_index` facilitating visualizations.

2.3 Analysis of the L–time diagram

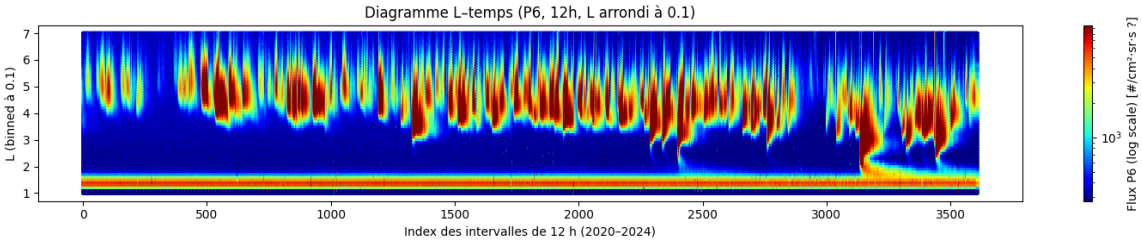


Figure 1: *L–time diagram showing the evolution of* 1 MeV *electron fluxes over the period 2020–2024. The colour scale represents the logarithm of the flux.*

Figure 1 shows an *L–time* diagram (L-shell versus time) displaying the average intensity of the 1 MeV electron flux (P6 channel) measured by the NOAA-15 satellite over the period 2020–2024.

The vertical axis represents the L parameter (`L_IGRF`), a magnetic coordinate characterizing the drift shell of particles (the larger L is, the farther from the Earth). The horizontal axis represents time, discretized into 12 hour intervals.

2.3.1 Key observations

Structure of the radiation belts. A high and relatively stable electron flux (warm colours: yellow/red) is observed around $L \approx 3$ and $L \approx 6.5$. The lower layer ($L \approx 1.2$) corresponds to the inner radiation belt, where electrons are trapped by the Earth's magnetic field. The upper layer ($L \gtrsim 4.5$), with similarly high fluxes, corresponds to the outer radiation belt.

Dynamics during geomagnetic storms. During the most intense storms (for example around time indices $\sim$ 2400 or $\sim$ 3200), electrons reach lower L shells than usual. Under quiet conditions, the "slot" or low-flux region lies around $L \approx 2.5$. During storms, the strong fluxes of the outer belt temporarily extend towards smaller L values (down to $L \approx 2.0$ or sometimes 1.8). This inward motion (radial diffusion of electrons) is an important indicator of the disturbance of the near-Earth space environment by solar activity.

3 Statistical analysis

3.1 Flux distribution and event definition

The distribution of the P6 flux is highly skewed, with a majority of low values and a few very high-flux episodes. To define *events*, the 95th percentile was computed:

$$\boxed{q_{0.95} = 11103.31 \quad (\log_{10}(q_{0.95}) \approx 4.05)}$$

The binary variable `event` is defined as:

$$\texttt{event} = \begin{cases} 1, & \text{if flux} > 11103.31 \\ 0, & \text{otherwise} \end{cases}$$

This threshold identifies the 5 % of measurements corresponding to the most intense fluxes, characteristic of periods of high geomagnetic activity.

3.2 Main visualizations

Three types of visualizations were produced to characterize flux variability:

- **Time series**: evolution of the P6 flux over a day, showing rapid orbital variations linked to the satellite crossing different regions of the magnetosphere.
- **L-shell trajectory**: variation of the magnetic L-shell over time, illustrating the polar orbit of the satellite.
- **L–time diagram with events** (Figure 2): 2D representation (L-shell × time) with flux in logarithmic scale, revealing the structure of the radiation belt.

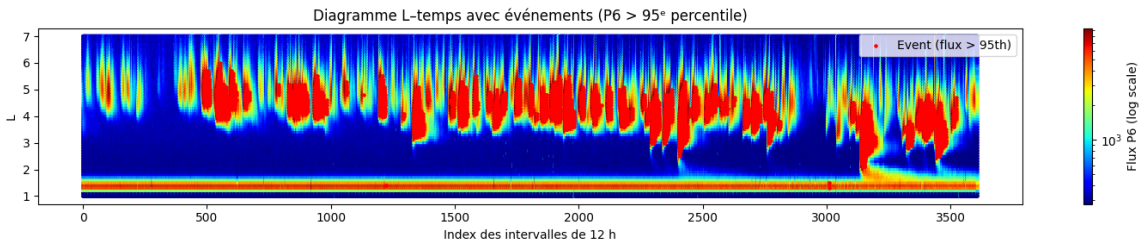


Figure 2: *L–time diagram (2020–2024) with events (flux > 95th percentile) marked in red.*

Figure 2 shows that events (red points) are concentrated around $L \approx 4$, consistent with the dynamics of the outer belt during periods of high geomagnetic activity.

4 Integration of OMNI data and preparation for modelling

This section corresponds to the second phase of the project, where solar wind parameters are integrated to improve the predictive capability of the models.

4.1 Acquisition and selection of OMNI variables

We used the `sunpy` library to download OMNI data at hourly resolution (dataset `OMNI2_HO_MRG1HR`). The variables selected for modelling are:

- **V**: solar wind speed.
- **BZ_GSM**: B_z component of the interplanetary magnetic field (GSM coordinates).
- **F10_INDEX**: solar radio flux at 10.7 cm.

4.2 Cleaning of OMNI data

4.2.1 Handling of missing values (NaN)

An analysis of missing values showed a low percentage (maximum **1.49**% in **V**). As the distribution of NaNs is not cyclical, we removed all rows containing missing values (`dropna()`).

Variable	Missing values	%
V	654	1.49
BZ_GSM	351	0.80
F10_INDEX	24	0.05

Table 1: *Missing-value rate before cleaning.*

4.2.2 Physical outlier filtering

Filters based on realistic physical limits were applied to remove outliers:

- **V**: 200 to 750 km/s
- **BZ_GSM**: −30 to 30 nT
- **F10_INDEX**: 50 to 300 s.f.u.

This process removed **248** rows considered as outliers.

4.3 Aggregation and scaling

In line with the +12h prediction objective, hourly data (Omni and NOAA) were resampled by taking the mean over a period of **12** hours (`resample('12H').mean()`).

A **Z-score standardisation** (`StandardScaler`) was applied to **V** and **BZ_GSM** to normalise their distributions before regression.

5 Regression modelling (Flux prediction)

The objective is to predict the $\log_{10}$ of the electron flux (log_flux) at **+12h**.

5.1 Features and preparation for regression

- **Independent variables (X)**: V, BZ_GSM, F10_INDEX, and log_flux_12h_ago (previous flux value).
- **Target variable (y)**: log_flux (shifted by $+12h$).

The data are split **chronologically** (60% Training, 20% Validation, 20% Test) for each L-shell ($2.8 \leq L \leq 7.0$).

5.2 Model training per L-shell

A **linear regression model** (`LinearRegression`) is trained within a standardised pipeline for each L-shell.

5.3 Prediction example ($L = 4.0$)

The model obtained a training-set R^2 score of **0.865** for the $L = 4.0$ shell.

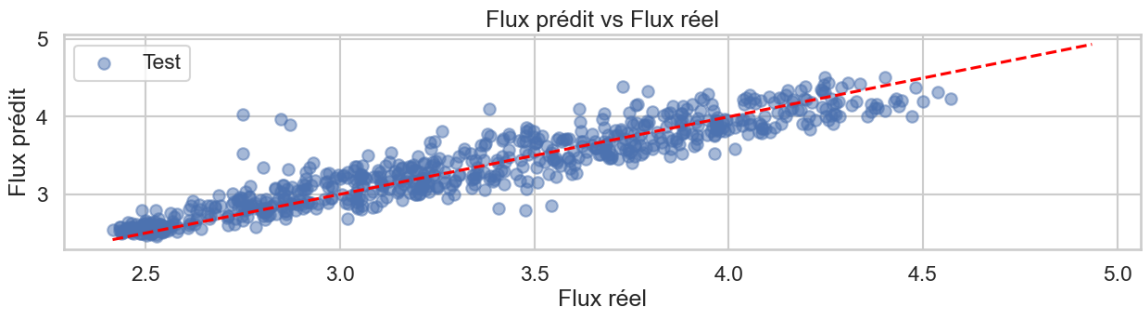


Figure 3: *Comparison between predictions (y_{test_pred}) and true values (y_{test}) for the $L = 4.0$ shell.*

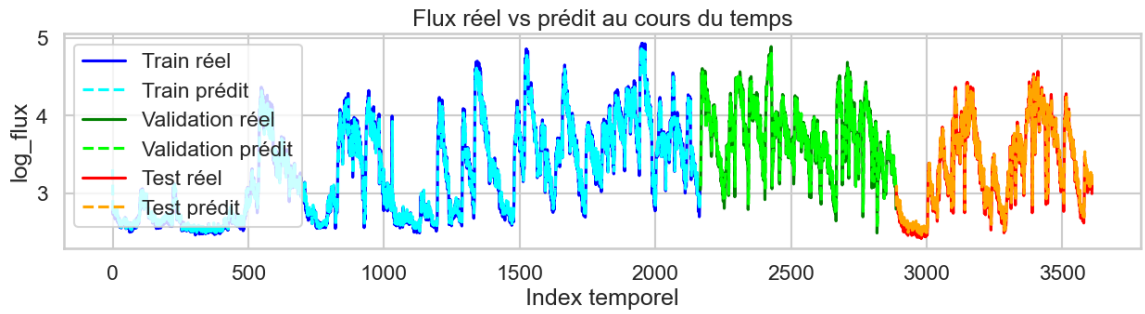


Figure 4: *Comparison of predictions and observations over time for the $L = 4.0$ shell.*

6 Results and discussion

6.1 Performance of the regression model

The linear regression models trained for each L-shell show varying performances:

- The best performances ($R^2 > 0.85$) are observed around $L = 4.0$, corresponding to the core of the outer belt.
- Performance decreases at the extremes ($L < 3$ and $L > 6.5$), reflecting greater variability and lower fluxes in these regions.
- The inclusion of the `log_flux_12h_ago` variable is crucial, indicating strong temporal persistence of electron fluxes.

6.2 Limitations and perspectives

6.2.1 Identified limitations

- The linear model does not capture complex nonlinearities during extreme events.

- The 12-hour temporal resolution may mask rapid variations.
- The absence of other geomagnetic indices (Kp, Dst) limits the full characterisation of magnetospheric activity.

6.2.2 Future improvements

- Implementation of nonlinear models (Random Forest, Gradient Boosting, neural networks).
- Extension to longer prediction horizons (+24h, +48h).
- Integration of additional geomagnetic indices.
- Development of a classification model for event detection.

7 Conclusion

This project led to the development of a complete pipeline for the prediction of 1 MeV electron fluxes in low Earth orbit:

1. **Acquisition and preprocessing**: robust processing of 5 years of NOAA-15 and OMNI data with physical and statistical filtering.
2. **Exploratory analysis**: identification of the structure of the radiation belts and of high-flux events.
3. **Modelling**: development of regression models per L-shell with satisfactory performance ($R^2 \approx 0.87$ for $L = 4.0$).

The results demonstrate the feasibility of predicting electron fluxes at +12h by combining satellite measurements with solar wind parameters. This work provides a solid basis for future developments towards operational space-radiation warning systems.